Title of Topic: Three Level DBMS Architecture

Course Code: 23IT201

Course Title: DataBase Management Systems

Session Number: 2

Academic Year: 2024 - 2025 (Even Sem)

MODULE 1-INTRODUCTION

1. Introduction to DBMS, Characteristics of DBMS, DBMS vs File Systems, need for DBMS.
2. Three Level DBMS Architecture
3. Data Models Introduction, Benefits, and Phases
4. ER Diagrams – Symbols, Components, Relationships, Weak entities, Attributes, Cardinality.
5. Relational Algebra, Domain Relational Calculus, Tuple Relational
6.Calculus Keys - primary Key, Foreign Key.

Three Level DBMS Architecture

Topics to be Covered

- Data Abstraction
- Levels of Data Abstraction
- Data Independence
- Types of Data Independence
- Data Independence in File System vs DataBase System
- DBMS Architecture
- DataBase System Architecture

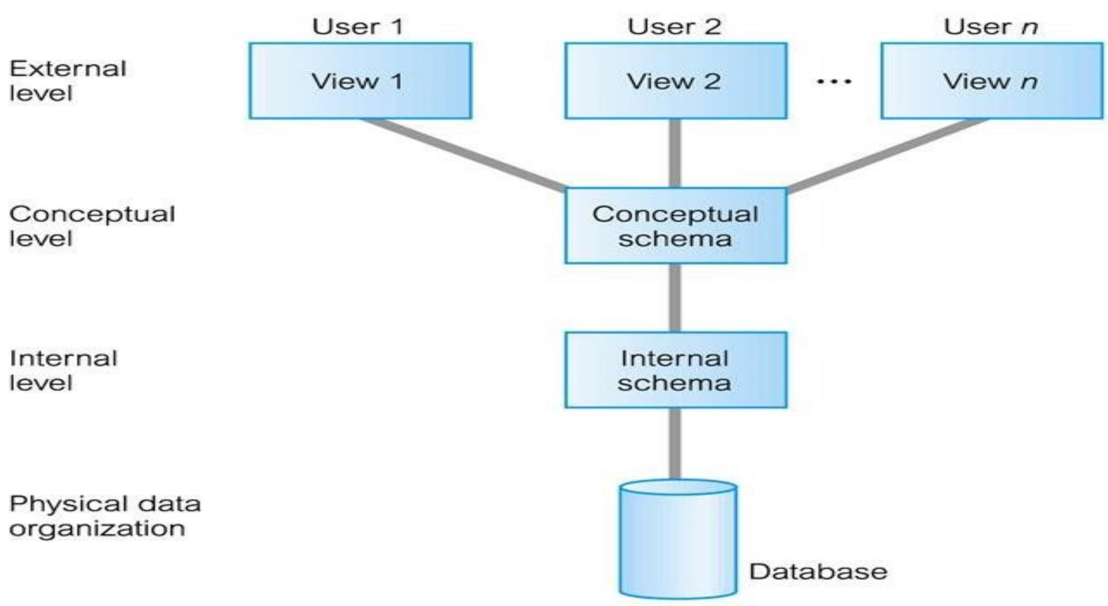
Data Abstraction

- The process of hiding certain details of database from user
- Hides some details like how data is stored and created Provides only the required data
- A database system that is able to separate the three different views is likely to be flexible and adaptable
- This flexibility and adaptability is called data abstraction

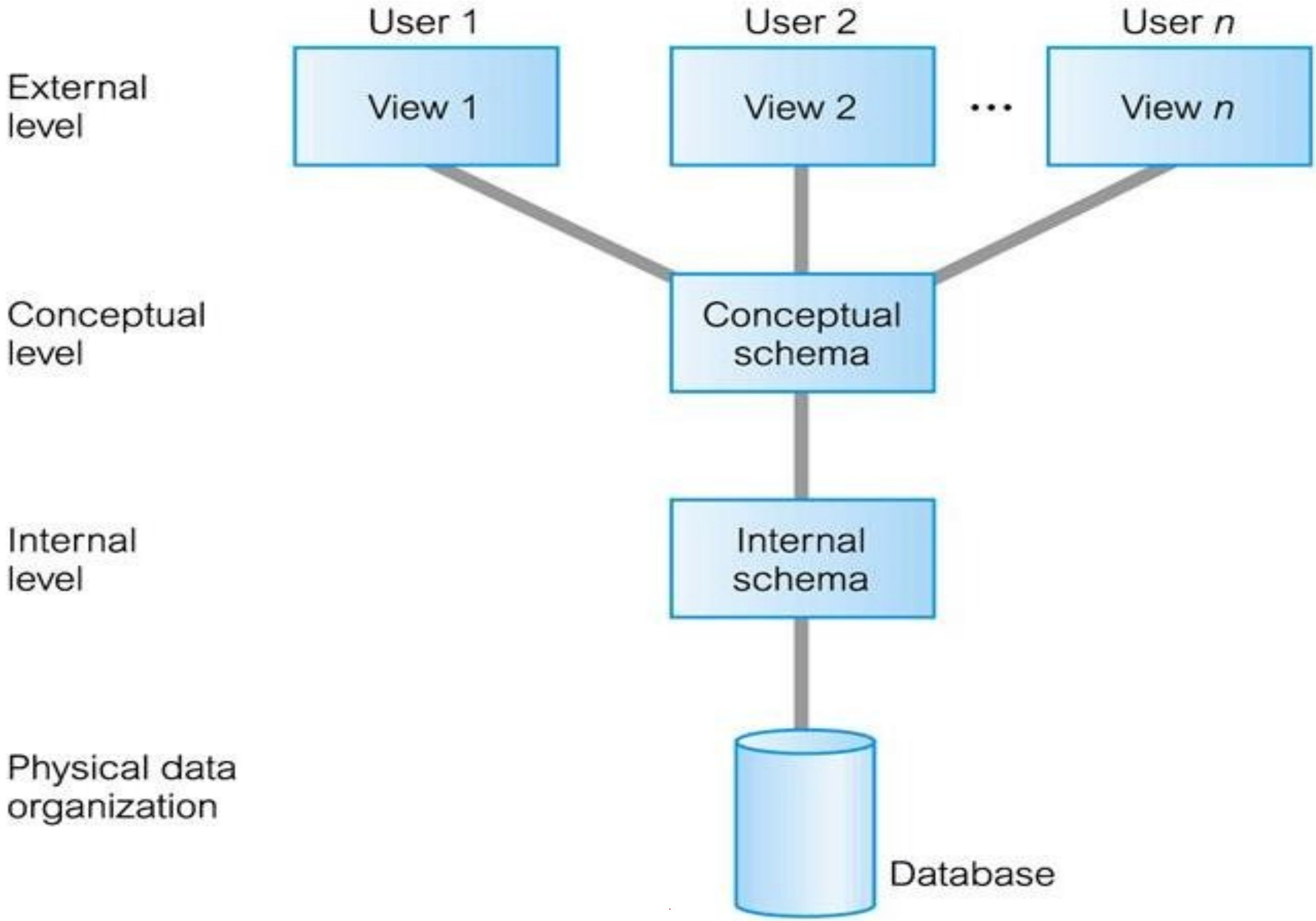
Data Abstraction:

Levels of Data Abstraction

- Architecture based on user view (or) functions
- ANSI/SPARC produced an architectural framework for databases with three different views of the data
- Used for describing the structure of specific database systems



Architecture:


User 1
User 2
User n
External level
View 1
View 2
...
View n
Conceptual level
Conceptual schema
Internal level
Internal schema
Physical data organization
Database

Levels of Data Abstraction

Internal/Physical View at Internal/Physical Level

- Lowest level of data abstraction which has the knowledge about s/w and h/w
- Describes what data is stored in the database system and how
- Describes complex low-level data structures in detail
- Keeps the information about the actual representation of the entire database
- Closest to the physical storage and does not deal with the physical devices directly
- Instead it views a physical device as a collection of physical pages and allocates space in terms of logical pages
- Also considers file organization, storage allocation, access paths of indexes, data compression and encryption techniques and record placement

Levels of Data Abstraction

Conceptual View at Logical Level

- Next higher level of abstraction (Inbetween user level and physical level)
- Describes what data are stored in the database and what relationship exist among these data
- Describes entire database in terms of small number of relatively simple structures and more stable than other two views
- Hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations and constraints
- Consists of conceptual schema which can be changed without changing anything in the application

Levels of Data Abstraction

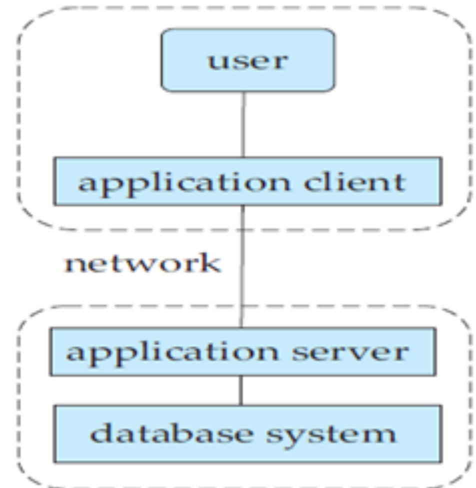
External View at External Level

- Highest or Top level of data abstraction (No knowledge of DBMS s/w, h/w or physical storage) concerned with the user
- Describes only part of the entire database that a particular user i interested in and hides the rest of the database from user
- There can be n number of external views for database where n is the number of users
- **E.g.,** An accounts department is only interested in the student fee details. It does not have any interest in the personal information about students
- All the database users work on external level of DBMS
- Views of individual users are defined by means of external schemas

DBMS Architecture

3-Tier Architecture

- Application program runs on client side and Database is stored on server



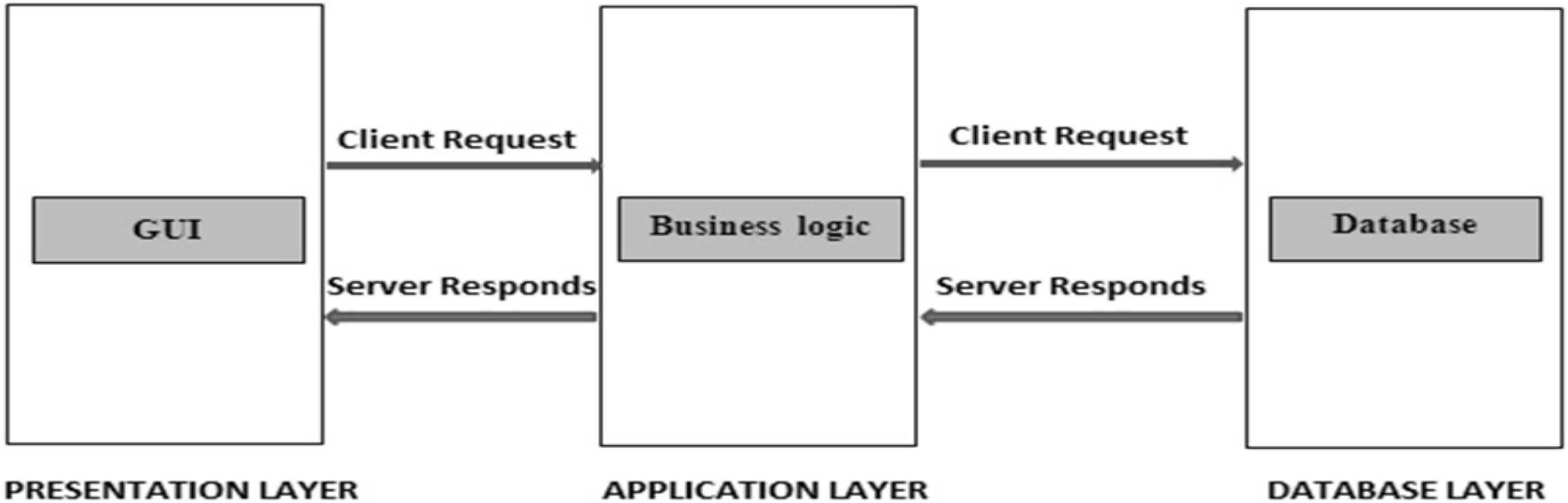
n a separate server

- This layer stores the programs that access the database from database server

DBMS Architecture

3-Tier Architecture

- Application layer presents the data to the user in a required form
- Three layer work together to create 3-Tier Architecture
- Each of the layer can be operated and managed individually
- **E.g.,** Web Application such as E-Commerce



3 – Tier Architecture

DBMS Architecture

N-Tier Architecture

- Used in web based application with three layers/tiers
- N-Tier architecture begins as a 3-Tier model
- **E.g.,** Any large website on the internet

DBMS Architecture

Middle Tier (Application Layer)

- Contains business logic and communicates between the client and server
- Holds the application server and the programs that access the database
- End users are unaware of any existence of the database beyond the application tier and not aware of any other user beyond the application tier
- Also processes functional logic, constraint, and rules before passing data to the user or down to the DBMS

Data Tier

- Manages the database
- Databases resides along with its query processing languages and the respective relations

DBMS Architecture

Presentation Tier (Client Layer)

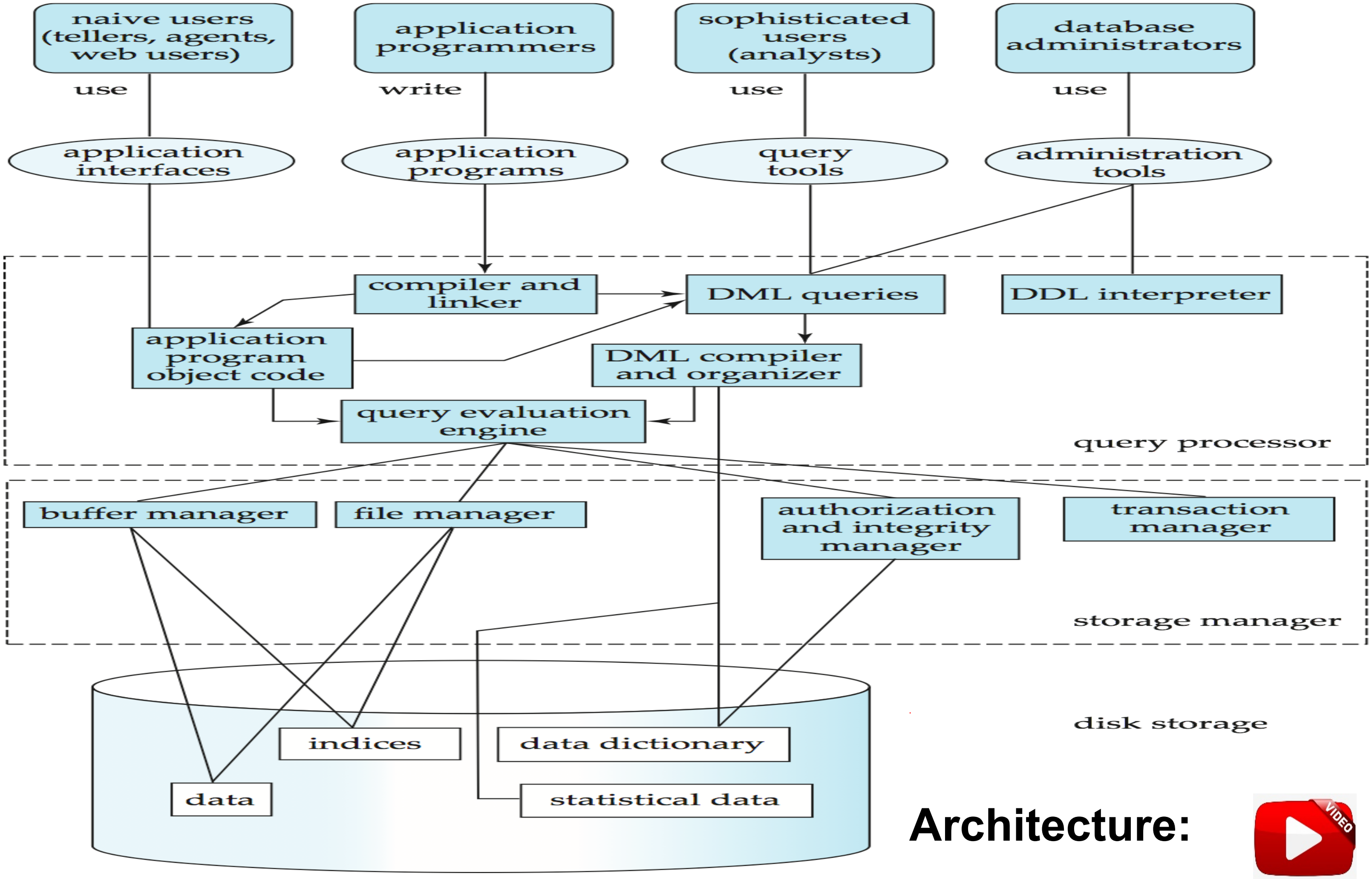
- Interface between the user and the system
- Contains GUI part of the system (your PC, Tablet, Mobile, etc…)
- All end users operate on this tier
- Multiple views of the database can be provided by the application
- All views are generated by the application that reside in the application tier

Advantages

- Easy to maintain and modify without affecting other modules
- Improved Security as client cannot access the database easily
- Improved Performance and Data Integrity, also Fast Communication

DataBase System Architecture

- Allows communication between different types of user with physical DB and OS



Architecture:

Database System Architecture

User Communication, Authentication and Authorization

- Deals with the user when he/she communicates with the DBMS
- Communication depends on the kind of DBMS (2-TIER or 3-tier)
- User first log on to the system and has to be authenticated
- Once the user is allowed to access the database, the system will check the user's credentials
- Naive users use application interfaces
- Application programmers write application programs
- Sophisticated users use query tools
- Database Administrators use administration tools

DataBase System Architecture

Query Processor (Executes of DDL and DML statements)

- Application program object code - Directly connected with application interface
- Compiler - Connected with application program will compile them
- DML Queries - Connected with query tools
- DDL Interpreter – Interprets DDL statements into a set of tables containing metadata
- DML Compiler and Organizer – Translates DML statements into low level instructions that the query evaluation engine understands
- Query Evaluation Engine – Executes low level instructions generated by DML compiler

DataBase System Architecture

Storage Manager

- Provides interface between low-level data stored and application program or queries
- Transaction Manager – Simply preserves atomicity (all or none) and controls concurrency (coordination of the simultaneous execution of transactions in a multiprocessing database)
- Authorization and Integrity Manager – Checks the authority of users to access data and integrity constraints
- File Manager – Manages allocation of space on disk storage
- Buffer Manager – Fetches data from disk storage to memory for being used

DataBase System Architecture

Disk Storage

- Where the actual data is stored
- Data Dictionary – Store metadata (data about data)
- Indices – Provide faster access to data items
- Data – Store user data (Nothing but warehouse)
- Statistical Data – Store statistical information about the data

Three Level DBMS Architecture & Data Abstraction, Data Independence, DB System Architecture

Summary

- Data Abstraction
- Levels of Data Abstraction
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Three Level DBMS Architecture & Data Abstraction, Data Independence, DB System Architecture

Try Yourself

- A learner of DBMS store and manipulate the data on the same system. Find the architecture used in this system.
- If you have used MySQL, then you must have seen PHPMyAdmin. Identify which type of architecture is followed in this system.
- Identify the architecture used in Desktop Games.
- Draw the three-level data abstraction for

✓ Customer Database with the attributes such as Cust_ID, F_Name, L_Name, Contact Number, SSN, City.

✓ Employee Database with the attributes such as Empl_No, F_Name, L_Name, Age, Salary, DoB, Branch_No.

Session 3 - Introductions to Data Models-Hierarchical Model, Network Model, Object Oriented Model, Entity–Relationship Model, Relational Model